Lab "Normalization"

① تكرار البيانات بسبب Projectid وبالتالي Primary Key

Eid	Ename	Eage	Eaddress
1	Ahmed	20	bahna
2	Ali	25	Cairo
5	Mona	24	Alex
15	Khalad	32	Cairo

1 Normalization Form

Composite Key

Eid	Projectid	Projectname	Locationid	Location	hour	Bouns
1	15	[Marsso, Matrooh airPort]	137	Matrooh	243	40000
1	7	[hebata, airPort]	137	Matrooh	193	30000
2	6 *	[Proj ALArab airPort]	200 *	Alex *	210	32000
5	6 *	Proj ALArab airPort *	200 *	Alex *	150	21000
5	15	Marssa Matroch airPort	137	matrooh	130	19000
5	23 →	Cair airPort →	55 →	Cairo →	537	56000
15	23 →	Cairo → airPort	55 →	Cairo →	33	5000

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② بدرع Composit Key عشان اشوف هي Composit Key

Partial dependency

- Projectname → Projectid
- Projectid Locationid → Projectid
- Location → Projectid
- hour → Eid, Projectid
- Bonus → Eid, Projectid

توضيح الـ Partial dep

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Eid	Projectid	hour	Bonus
1	15	243	40000
1	7	193	3000
2	6	210	32000
5	6	150	21000
5	15	130	9000
5	23	537	56000
15	23	33	5000

ProId	Projectname	Locationid	Location
15	Marssa Matrooh airPort	137	Matrooh
7	hebata airbort	137	Matrooh
6	Proj-ALArab airPort	200	Alex
23	Cairo airPort	55	Cairo

Relence

③ حذف اوفيل

Project ID	Project name	Location id
15	Marssa Matrooh airPort	137
7	hebta airPort	137
6	ALArab airPort	200
23	Cairo AirPort	55

Location id	Location
137	Matrooh
200	Alex
55	Cairo